[CS3809]: SPRINT [1] REPORT (TEAM #[4])

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1. Introduction

BudgetWise is a web-based application designed to help students and employees manage their monthly income and expenses. It allows users to divide their income into Needs, Wants, and Savings using the 50/30/20 rule or customized percentages. The system helps users track expenses, set savings goals, and avoid overspending through a simple and user-friendly interface. The project will be developed using Agile Scrum methodology across three sprints.

1.1 Project Objectives and Problem Statement

Many students and employees face difficulties managing their monthly income due to unorganized spending habits, poor expense tracking, and a lack of clear financial planning. As a result, they may spend more than their available budget, struggle to save money, or fail to achieve their financial goals. Managing expenses manually can also be time-consuming and make it difficult to understand spending patterns.

1.2 Product Vision (Moore's Vision Template)

Budget Wise aims to give students and employees one simple place to plan their monthly income, record their expenses, and see how much money they have left, so they can spend with confidence and reach their savings goals.

Complete the following statement to define your product vision:

For	People who want to manage their personal money.
Who	Need an easy way to track their income, expenses, and spending plans.
The	Budget Wise
Is a	Personal budget management system.
That	Helps users record their expenses, manage their budgets, and see how much money they have left.
Unlike	Using notebooks or separate spreadsheets to track money.
Our product	Provides one organized place to divide income into Needs, Wants, and Savings, track spending against the budget, and view spending summaries.

1.3 Methodology (Agile)

We chose the Agile methodology for our Budget Wise project because it makes the development process more flexible and allows us to improve the system step by step. Unlike the traditional Waterfall

approach, Agile allows the team to make changes during development and receive feedback regularly. This helps us find problems early and improve the system throughout the project.

In each sprint, the team goes through five main phases:
Planning: Set the sprint goals, select the user stories to implement, and divide the tasks among team members.
Design: Prepare the system diagrams, database design, and user interface layouts needed for the sprint.
Development: Build the selected features step by step.
Testing: Test the new features to make sure they work correctly, and fix any errors found.
Deployment: Prepare the working version of the system so it is ready for review and use.

1.4 Sprint schedule

Sprint	Main Focus	Duration
Sprint 1	Requirements Analysis, User Registration and Login, Monthly Income Management, Basic UI Design, and Database Setup.	Week 11
Sprint 2	[e.g., Core feature development]	Week
Sprint 3	[e.g., Testing, refinement, and deployment]	Week 14

2. Requirements

The requirements of BudgetWise were identified based on the needs of students and employees who want to manage their monthly income and expenses. The system aims to help users organize their budgets, track their spending, and achieve their savings goals.

For Sprint 1, five user stories were selected with a total of 13 User Story Points (USP). These stories focus on the basic functions of the system.

2.1 Backlog

List the user stories planned for this sprint:

ID	Name	USP	Priority
Issue-[01]	*As a user, I want to register and log in so that I can securely access my account.*	*[2]*	*[1]*
Issue-[02]	*As a user, I want to enter my monthly income so that I can manage my available budget.*	*[2]*	*[2]*
Issue-[03]	*As a user, I want to allocate my income into needs, wants and savings so that I can organize my money.*	*[3]*	*[3]*
Issue-[04]	*As a user, I want to record my daily expenses so that I can track*	*[3]*	*[4]*

	my spending.		
Issue-[05]	*As a user, I want to view my financial dashboard so that I can monitor my budget and expenses.*	*[3]*	*[5]*
Total		*[13]*	

Priority 1 represents the highest priority.

2.2 User Stories

Detail each user story listed in the backlog above using the format below (repeat for every story):

ID	*Issue-[01]* *User Registration and login*
As a	*User*
I want to	*Register and login to the system.*
So that I can	*Securely access my personal account.*
USP	*[2]*
Priority	*[1]*
Description	*The System allows new users to create an account and existing users to login using their credentials.*

ID	*Issue-[02]* *Monthly income management*
As a	*User*
I want to	*Enter my monthly income*
So that I can	*Manage my available money*
USP	*[2]*
Priority	*[2]*
Description	*The system allows users to enter their monthly income, whether it comes from a salary, university allowance, or personal budget.*

ID	*Issue-[03]* *Budget Allocation*
As a	*User*
I want to	*Divide my income into needs, wants and savings.*
So that I can	*Organizing my spending and save money.*
USP	*[3]*
Priority	*[3]*
Description	*The system allows users to divide their income using the 50/30/20 budget rule or customize the percentages according to their percentages.*

ID	*Issue-[04]* *Daily Expense Tracking*
As a	*User*
I want to	*Record my daily expenses.*
So that I can	*Track my spending and avoid over spending.*
USP	*[3]*
Priority	*[4]*
Description	*The system allows users to record daily expenses and assign them to appropriate budget categories.*

ID	*Issue-[05]* *Financial dashboard*
As a	*User*
I want to	*View my financial dashboard.*
So that I can	*Monitor my income expenses and remaining budget.*
USP	*[3]*
Priority	*[5]*
Description	*The system displays an overview of the user's monthly income, budget, allocation, and record expenses.*

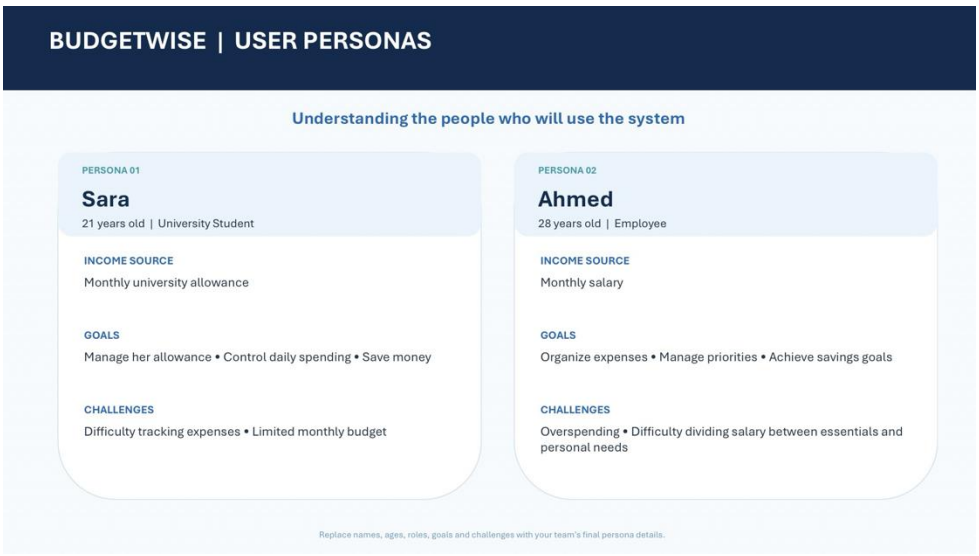
2.3 User Personas

Persona 1: university student

- ***Name:*** *Sara*
- ***Age:*** *21*
- ***Occupation:*** *University Student*
- ***Income Source:*** *Monthly university allowance*
- ***Goals:*** *Manage her allowance, control daily spending, and save money.*
- ***Challenges:*** *Difficulty tracking expenses and managing her limited monthly budget.*

Persona 2: Employee

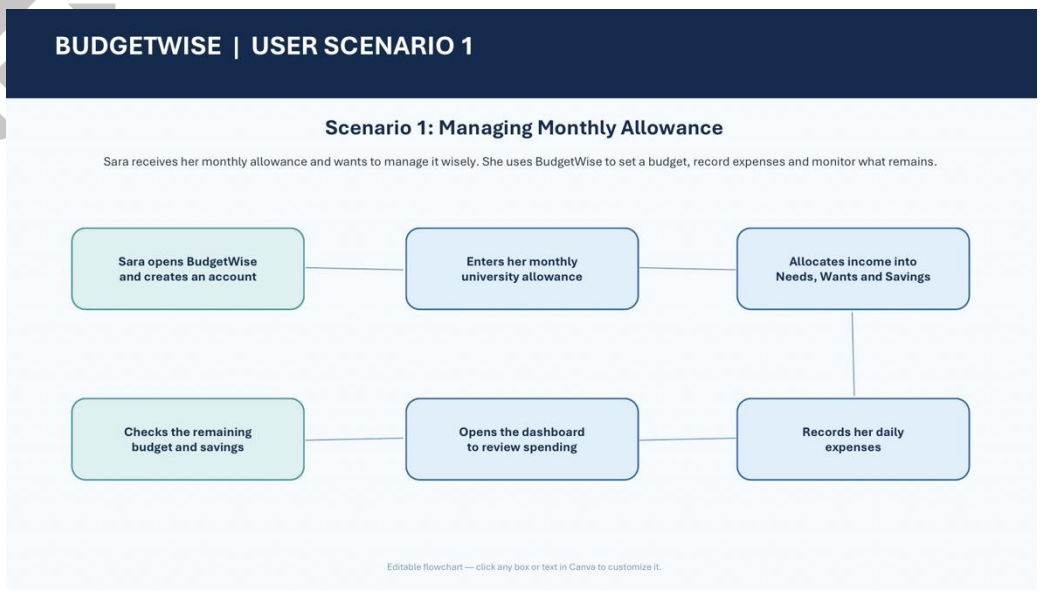
- ***Name:*** *Ahmed*
- ***Age:*** *28*
- ***Occupation:*** *Employee*
- ***Income Source:*** *Monthly Salary*
- ***Goals:*** *Organize monthly expenses, manage financial priorities, and achieve savings goals.*
- ***Challenges:*** *Overspending and difficulty dividing his salary between essential expenses and personal needs.*



2.4 User scenarios

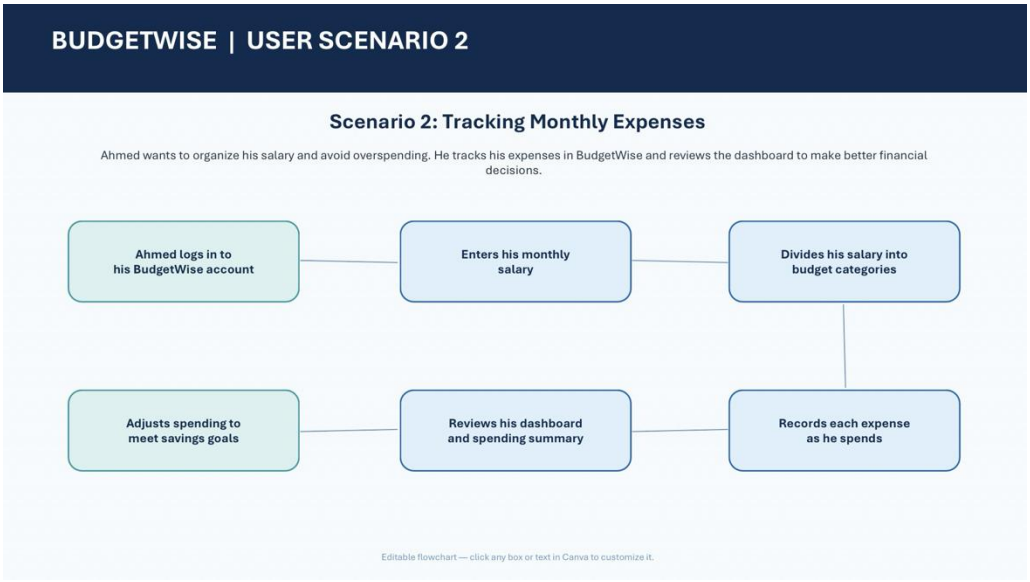
Scenario 1: Managing Monthly Allowance

Sara is a university student who receives a monthly allowance. She creates an account in BudgetWise and enters her monthly income. She then divides her money into Needs, Wants, and Savings using the 50/30/20 rule. Throughout the month, she records her daily expenses and checks her dashboard to monitor her remaining budget.



Scenario 2: Tracking Monthly Expenses

Ahmed is an employee who wants to organize his monthly salary. He logs in to BudgetWise, enters his salary, and allocates it into different budget categories. He records his expenses regularly and uses the financial dashboard to review his spending and make better financial decisions.



** Note: Repeat the Issue / USP / Priority / Description block above for every user story in your backlog.*

3. Release Planning

3.1 Sprint [1] Summary

Sprint 1 focused on requirements analysis and establishing the basic structure of the BudgetWise system. The planned work included user registration and login, monthly income management, basic user interface design, and database setup.

Five user stories were selected for Sprint 1, with a total of 13 User Story Points (USP). These stories covered user registration, monthly income management, budget allocation, daily expense tracking, and the financial dashboard.

During this sprint, the team documented the system requirements, identified the main user stories, and developed user personas and scenarios to understand user needs. The completion status of the planned features and any challenges encountered during development will be confirmed based on the team's sprint review.

Summarize the status of each user story planned for this sprint:

Story ID	Story Title	USP	Status
Issue-[01]	User Registration and Login	2	*DONE*
Issue-[02]	Monthly Income Management	2	*DONE*
Issue-[03]	Budget Allocation	*3*	*DONE*
Issue-[04]	Daily Expense Tracking	*3*	*DONE*
Issue-[05]	Financial Dashboard	*3*	*DONE*
Total		13	

3.2 Sprint [2] Planning (omit in your final sprint report)

Sprint 2 will focus on developing additional features of BudgetWise and improving the connection between the system's main functions. The proposed features include savings goals, budget alerts, and monthly expense reports.

Three new user stories are proposed for Sprint 2, with an estimated total of 9 User Story Points (USP), compared to 13 USP planned for Sprint 1. Any unfinished stories from Sprint 1 will also be reviewed and considered for the next sprint.

Possible risks include limited development time and technical difficulties when integrating new features with the existing system. To reduce these risks, the team will prioritize important tasks, divide responsibilities clearly, and test the features during development.

Story ID	Story Title	USP	Status
Issue-[06]	Savings Goals	3	*[Planned]*
Issue-[07]	Budget Alerts	3	*[Planned]*
Issue-[08]	Monthly Expense Reports	*3*	*[Planned]*
Total		9	

4. Architecture

4.1 High-Level Overview of System

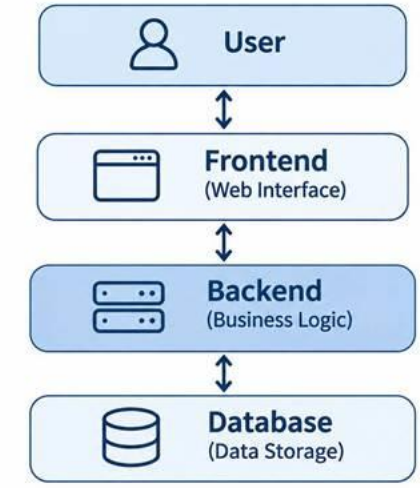
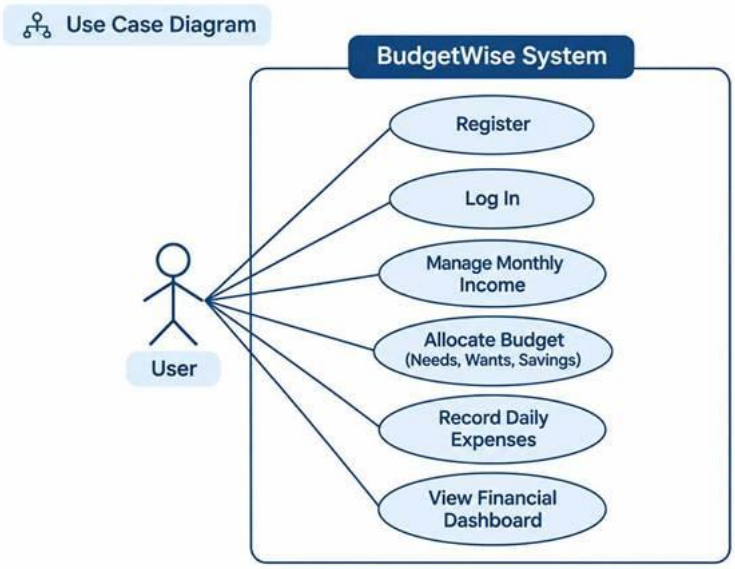


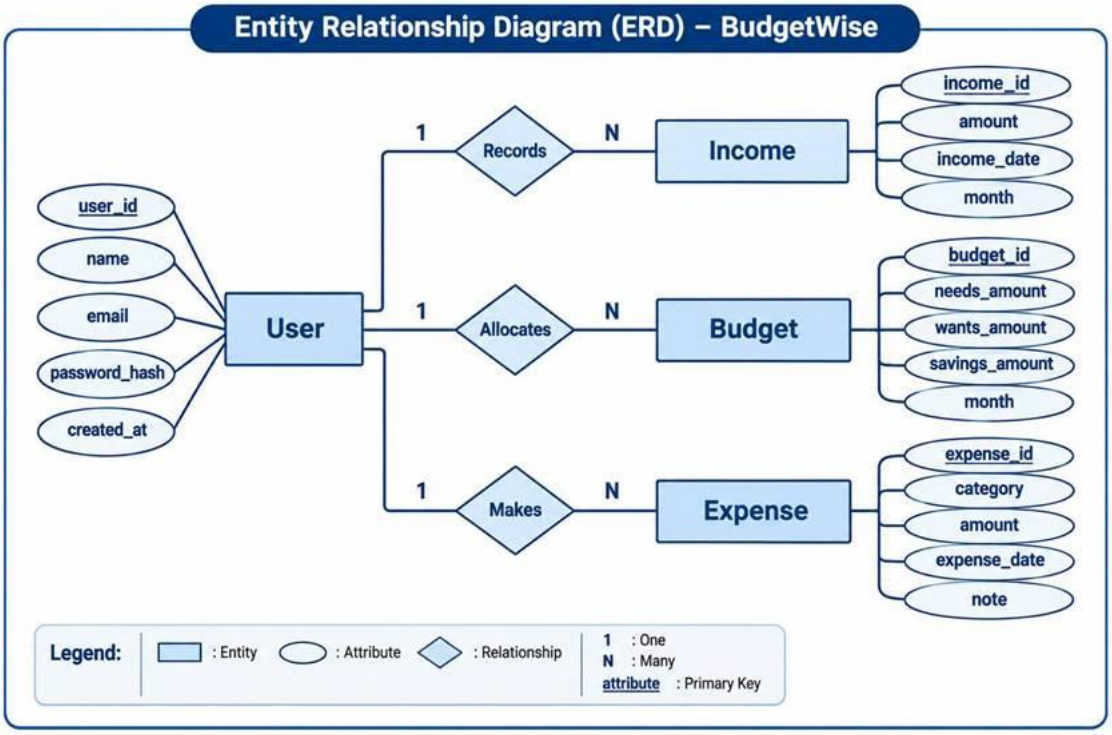
Figure 4.1: High-Level Overview of System

The figure below shows the proposed initial architecture of BudgetWise. The user interacts with the system via the frontend, where the user register, logs into the system, and manages there monthly income.The backend is where all user requests and budget operations are processed, and the database will contain all the user data and finance-related information. This architecture is initial and can change in the next sprints according to the project needs.

4.2 Use Case Diagram



5.4 Database Design



5. User Interface Design

Insert the UI mockups for each feature (user story) completed in this sprint, organized by user role/portal.

5.1 Sprint [N] UI Mockups

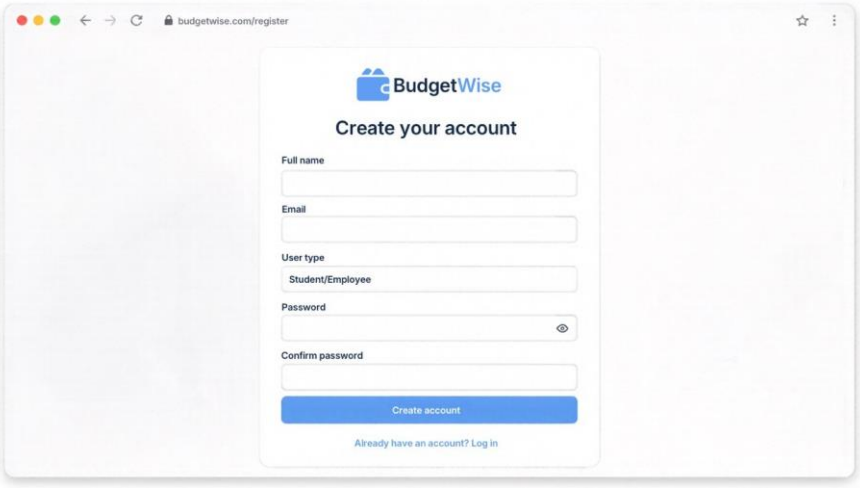


Figure 1: Registration screen for User Story Issue-01

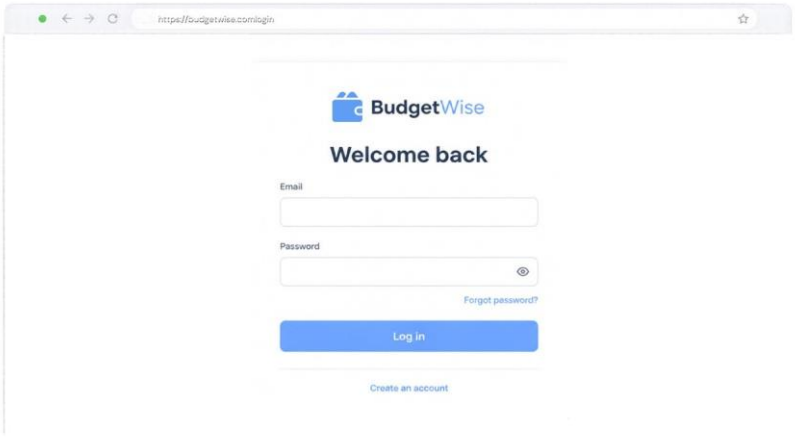


Figure 2: Login screen for User Story Issue-01

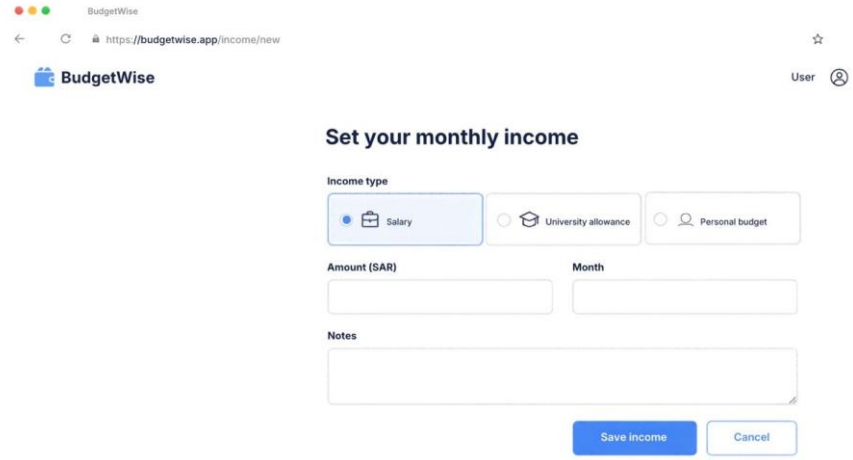


Figure 3: Monthly income entry screen for User Story Issue-02

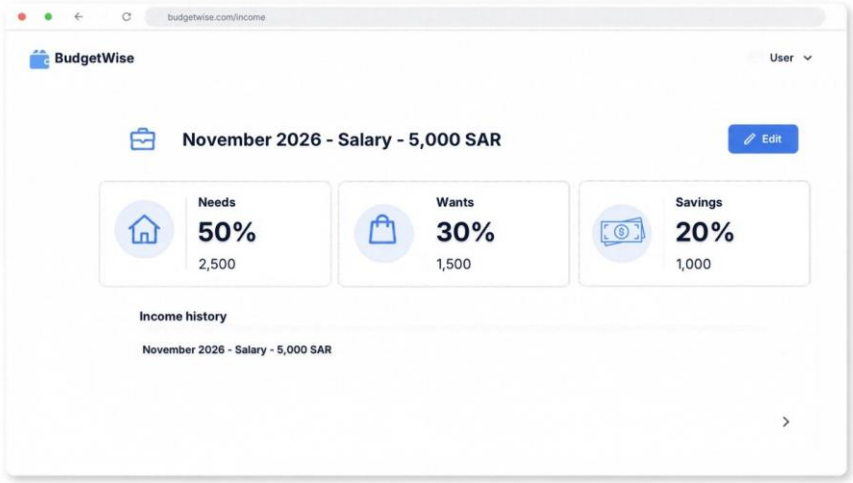


Figure 4: Income summary and edit screen for User Story Issue-02

6.1 Test Cases

Document each test case using the format below. Add as many as needed, organized by sprint or by user role.

Test case: TC-01 – Create a new account

User story: Issue-01

Pre-conditions: *The user does not have an account and is on the Create your account page.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Open the Create your account page. 2. Enter the full name and email. 3. Choose the user type (Student/Employee). 4. Enter the password and confirm it. 5. Click "Create account".	The account is created and the user can log in.	The account was created successfully.	Pass

Test case: TC-02 – Log in with correct data

User story: Issue-01

Pre-conditions: *The user has a registered account and is on the Welcome back page.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Open the Welcome back page. 2. Enter the registered email. 3. Enter the correct password. 4. Click "Log in"	The user is logged in and can access the personal account.	The user logged in and the account opened.	Pass

Test case: TC-03 – Enter the monthly income

User story: Issue-02

Pre-conditions: *The user is logged in and is on the Set your monthly income page.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Select the income type (Salary). 2. Enter the amount in SAR (5,000). 3. Select the month (November 2026). 4. Click "Save income".	The income is saved and shown in the Income history.	The income was saved and appeared in the Income history.	Pass

Test case: TC-04 – Allocate income using the 50/30/20 rule

User story: Issue-03

Pre-conditions: *The user is logged in and has saved a monthly income of 5,000 SAR.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Open the Income page. 2. Check the Needs, Wants, and Savings cards.	Needs 50% (2,500), Wants 30% (1,500), and Savings 20% (1,000).	The income was divided as expected and the three amounts were displayed correctly.	Pass

Test case: TC-05 – Record a daily expense

User story: Issue-04

Pre-conditions: *The user is logged in and has an income and budget allocation for the current month.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Open the expense page. 2. Enter the expense amount. 3. Choose the budget category (Needs, Wants, or Savings). 4. Save the expense.	The expense is saved and linked to the selected budget category.	The expense was saved under the selected category.	Pass

Test case: TC-06 – View the financial dashboard

User story: Issue-05

Pre-conditions: *The user is logged in and has income and recorded expenses.*

Test Steps	Expected Result	Actual Result	Status (Pass/Fail)
1. Log in. 2. Open the dashboard.	The dashboard shows the monthly income, the budget allocation, the recorded expenses, and the remaining budget.	The dashboard displayed all the information correctly.	Pass

7. Sprint [N] Retrospective

What went well:

In Sprint 1, we defined the main requirements of BudgetWise and created user personas and scenarios to understand our users better. We completed five user stories with a total of 13 story points. These included registration and login, monthly income management, budget allocation, daily expense tracking, and the financial dashboard.

What we will improve next:

In the next sprint, we will test new features early to make sure they work well with the existing system. We will also check our progress regularly and leave enough time to fix any problems before submission. We will update the report as we work to keep the documentation organized.

Our next steps:

In Sprint 2, we will focus on savings goals, budget alerts, and monthly expense reports. We will divide the tasks clearly and discuss any difficulties together to keep the work moving.

References

Flora, H. K., & Chande, S. V. (2014). A systematic study on agile software development methodologies and practices. *International Journal of Computer Science and Information Technologies, 5*(3), 3626–3637. https://www.ijcsit.com/docs/Volume%205/vol5issue03/ijcsit20140503214.pdf

Gaborov, M., Karuović, D., Radosav, D., Kavalić, M., Milosavljev, D., & Stanisavljev, S. (2021). The difference between Scrum and other agile and traditional methodologies. In E. Csanák (Ed.), *Proceedings of the 8th International Joint Conference on Environmental and Light Industry Technologies (IJCELIT 2021)*. Óbuda University. http://hdl.handle.net/20.500.14044/37740